NANA ADDO DANKWA BAMPOE ADDO

AI Engineer | LLM Applications & Workflow Automation

Passau, Germany | +49 176 7490 9253 | nanaaddoldoe@gmail.com

NanaAddoDBa | Nana-Addo | Available for full-time work immediately

PROFILE

Software developer specialising in LLM applications, retrieval-augmented generation and workflow automation. Combines JavaScript/TypeScript application development and Python/SQL automation with API integration, model-output validation and automated testing. M.Sc. Computer Science candidate at the University of Passau.

CORE TECHNICAL SKILLS

AI applications: LLM APIs, retrieval-augmented generation (RAG), prompt engineering, output validation, LangChain, OpenClaw

Automation: n8n, Microsoft Power Automate, SharePoint, Copilot Studio

Programming & APIs: Python, TypeScript, JavaScript, SQL, Node.js, REST APIs

Applications & data: React.js, Next.js, NestJS, PostgreSQL, Supabase, Prisma

Testing & delivery: Playwright, Cypress, Jest, Vitest, Git, Docker Compose, GitHub Actions (CI)

SELECTED PROJECTS

Guided Scratch Tutoring Assistant — 2026 – Present

Master's thesis | University of Passau | LLM APIs, LitterBox, Whisker, Docker Compose

- Built an LLM-powered Scratch tutor using learner project state, LitterBox static analysis and Whisker behavioural tests to generate context-aware guidance.
- Implemented validation gates for generated edits in isolated project copies; invalidated outdated guidance after workspace changes and required fresh tests before step completion.
- Verified the workflow with 102 passing focused tests across 10 suites and a three-step browser test using a live LLM; traced generation, validation and submission events for debugging.
- Packaged the local runtime in four Docker Compose services with health checks, separate provider configuration and persistent project storage.

AI Customer-Support Agent — 2026

OpenClaw | Retrieval-augmented generation (RAG) | LLM APIs | Telegram

- Developed an OpenClaw-based agent connecting LLM APIs, a client's business knowledge base and Telegram to answer customer enquiries using retrieval-augmented generation.

Expense Tracker & Budget Manager — 2026

Personal project | React.js, TypeScript, NestJS, Prisma, PostgreSQL

- Built a full-stack finance application with authenticated accounts, user-scoped expense records, budgets and savings goals.
- Implemented filtered, paginated REST APIs and configured frontend and API tests in GitHub Actions.

github.com/NanaAddoDBa/Mobile-expence-tracker

Flashcard Learning Platform — 2026

Personal project | React.js, JavaScript, Vite, Supabase, PostgreSQL

- Built a modular learning platform with authenticated deck libraries, persistent study sessions and recall-based review scheduling.
- Enforced session ownership, deck-version checks and duplicate-review prevention through server-side study functions.

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PROFESSIONAL EXPERIENCE

AI Automation and Marketing Analyst 10/2025 – 03/2026

AIESEC Germany | Passau, Germany | Volunteer

- Developed campaign reporting and email-generation workflows using Python, Power Automate and SharePoint, turning recurring data-processing tasks into repeatable automations.
- Analysed historical programme data to inform campaign targeting and enrolment planning.

Software Developer 2021 – 06/2025

Everything Beaded | Accra, Ghana | Remote

- Developed the company's e-commerce storefront in 2023 using React.js and Node.js, connecting customer-facing features with backend services.
- Built reusable frontend components and RESTful APIs, translating stakeholder requirements into web application features.
- Optimised database queries and application data retrieval to improve responsiveness across customer-facing workflows.

IT Officer (National Service) 09/2019 – 09/2020

Food and Drugs Authority (FDA) | Greater Accra Region, Ghana | Contract

- Developed internal auditing and verification tools to digitise manual regulatory reviews and support compliance tracking.
- Streamlined product-surveillance data collection, review and storage, preparing records for compliance checks and report generation.
- Built automated data workflows and product-metrics dashboards to support operational reporting.

IT Technician Intern 2018

Christian Community Microfinance Limited (CCML) | Accra, Ghana

- Developed SQL queries to extract client data and generate daily, monthly and annual reports for stakeholders.
- Resolved technical issues for financial-application users, supporting staff with day-to-day software problems.

EDUCATION & TECHNICAL QUALIFICATIONS

M.Sc. Computer Science 04/2023 – 03/2027 (expected)

University of Passau | Passau, Germany

B.Sc. Computer Science 06/2019

Christian Service University College, Kumasi | Awarded by the University of Ghana

Second Class Honours (Upper Division)

Advanced Professional Diploma in Database Technology 05/2014 – 09/2015

IPMC College of Technology | Ghana

Professional Diploma in Software Engineering 03/2014 – 03/2015

IPMC College of Technology | Ghana

SELECTED COURSES

Cypress: Web Automation Testing from Zero to Hero | Udemy 08/2024

React – The Complete Guide (incl. Next.js, Redux) | Udemy

Modern JavaScript From The Beginning 2.0 | Udemy

LANGUAGES

English: Fluent (C2) | German: A2 | Twi: Native